▼ VIEWPOINT

PREPARING INDIA FOR AN AI ECONOMY

While millions of job losses are expected, the new tech can enable the creation of many more high-impact jobs



BY B.V.R. SUBRAHMANYAM and DEBJANI GHOSH

AI is already reshaping jobs—coding, testing, customer support and data operations are among the first to feel the shift. Every technological revolution, from the printing press to computerisation, began with fear and ended with greater opportunity. AI will be no different. The real question is not if change will come, but how fast we can prepare for it.

Across India's $245 billion tech and customer experience (CX) sectors, automation is driving 15-20 per cent of productivity gains today, likely to rise to 50 per cent by 2030. Roles such as Level-1 support staff and junior developers will shrink. If we remain on a business-as-usual path, up to 2 million jobs may be displaced by 2031.

But this is only half the story. A new world of work is emerging—the once-standard categories of 'developer' or 'support engineer' are giving way to a web of hyper-specialised roles across three clusters. Firstly, professionals with Enterprise AI skills, meaning those who can deploy and manage AI systems in organisations. These include AI DevOps engineers, prompt designers and model auditors. The second cluster involves innovators with Frontier AI skills, those who are pushing the boundaries of science and technology, such as quantum-AI engineers, neuro-haptic designers and data ethicists. The third group is the researchers working on AI-for-AI skills, creating the next generation of intelligent systems: foundation-model developers, reinforcement-learning engineers and agentic-AI architects.

A key shift to recognise is that while the roles most at risk of displacement—such as data entry or routine testing—employed millions, each of the new AI-era roles may employ only hundreds or thousands today. However, taken together, they represent a far larger opportunity than the past. This is the inflection point for talent: the strategic pivot from fewer repetitive jobs to many more high-value, high-impact ones.



Illustration by RAJ VERMA

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This shift calls for a fundamental rethinking of how we build talent. The first priority is AI fluency for all: every student and worker must understand how to use AI responsibly. Equally important is interdisciplinary thinking: future roles will sit at the intersection of fields—like biology with data, or economics with ethics. We must train people to connect dots, not just memorise them. Another new reality is that skills now have short shelf lives, so continuous reskilling is the name of the game. Careers will depend on continuous learning, not one-time degrees. Finally, human-machine complementarity will be critical. As AI takes over logic-based tasks, the uniquely human skills of empathy, creativity and ethical reasoning will become our comparative advantage.

India has 9 million tech professionals—the world's largest digital talent pool—yet readiness for the AI economy remains uneven. Meanwhile, global competitors are not losing time—Saudi Arabia has trained 770,000 people in AI, Singapore funds 100 projects a year and the UAE runs the world's first AI university. India cannot afford delay.

If we act decisively, our tech and CX workforce could expand from 9 million today to 13 million by 2031, making India the global epicentre of AI talent and services. But this requires coherence. Multiple ministries and agencies are currently running isolated initiatives. What's missing is convergence, speed and shared accountability.

That's why NITI Aayog's Roadmap for Job Creation in the AI Economy calls for a unified India AI Talent Mission, which is a whole-of-government effort to equip, retain and empower India's AI workforce through three pillars, namely: embedding AI fluency across schools and universities, building a national AI skilling engine to reskill millions for new-age roles, and becoming a global AI talent magnet through research infrastructure, fellowships and incentives.

In tandem, this must align with the IndiaAI Mission, which provides the enabling layers of computing, datasets and open-source AI resources—turning trained talent into innovators. ■

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